Topic 2: Eigenvalues and Eigenvectors

Question 1:

Given the matrix

A = [[4, 1],
[2, 3]]

Write a Python program using `numpy.linalg.eig()` to:

1. Find the eigenvalues of A.
2. Find the corresponding eigenvectors.
3. Display each eigenvalue with its corresponding eigenvector.

Question 2:

Given the matrix

A = [[4, 1, 0],
[1, 4, 1],
[0, 1, 4]]

Write a Python program to:

1. Calculate all eigenvalues and eigenvectors.
2. Display the results.
3. Sort the eigenvalues in descending order along with their corresponding eigenvectors.

Question 3:

Given the matrix

A = [[2, 1, 0],
[0, 3, 1],
[0, 0, 4]]

Write a Python program to:

1. Find the eigenvalues and eigenvectors of A.
2. Form a matrix P using the eigenvectors as columns.
3. Form the diagonal matrix D containing the eigenvalues.
4. Verify the diagonalization relation A @ P ≈ P @ D